

VEEAM 100
UNDERGROUND

Architecting Veeam Backup for Microsoft365 v8 At Scale



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Sr. Product Infrastructure Architect

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See the Director's Cut!

Tuesday @ 11:15 AM

Pacific Ballroom 15



Architecting Veeam Backup for Microsoft365 v8 At Scale



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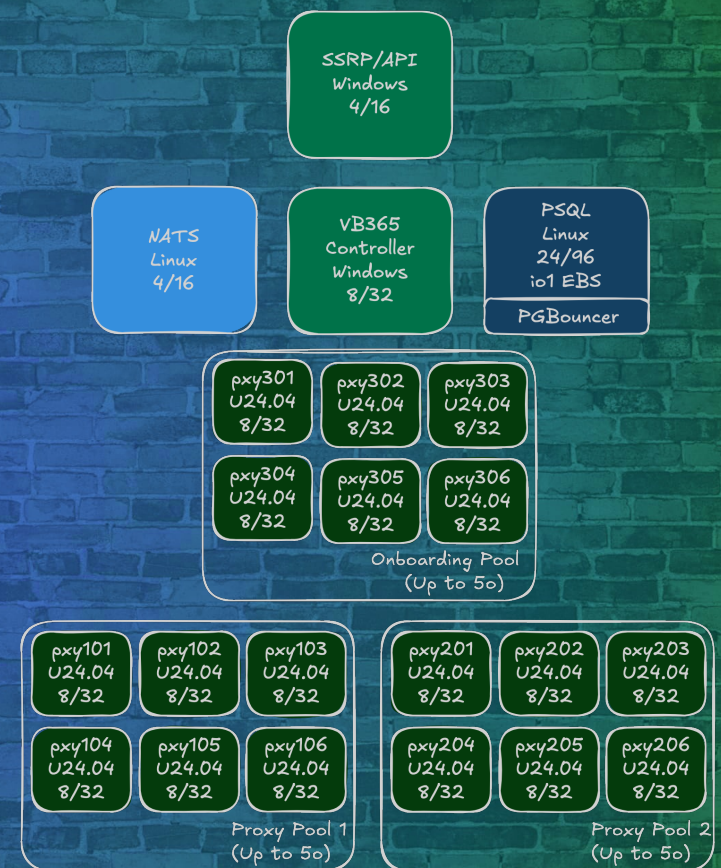
Designing for Scalability

- No all-in-one if you are big enough to be here
- Fully distributed architecture
- When possible, use Linux
- Cloud Always, compute close to storage
- Predict growth, both scale up and scale out
- Database, object storage compromise
- Address Exchange, Sharepoint components separately



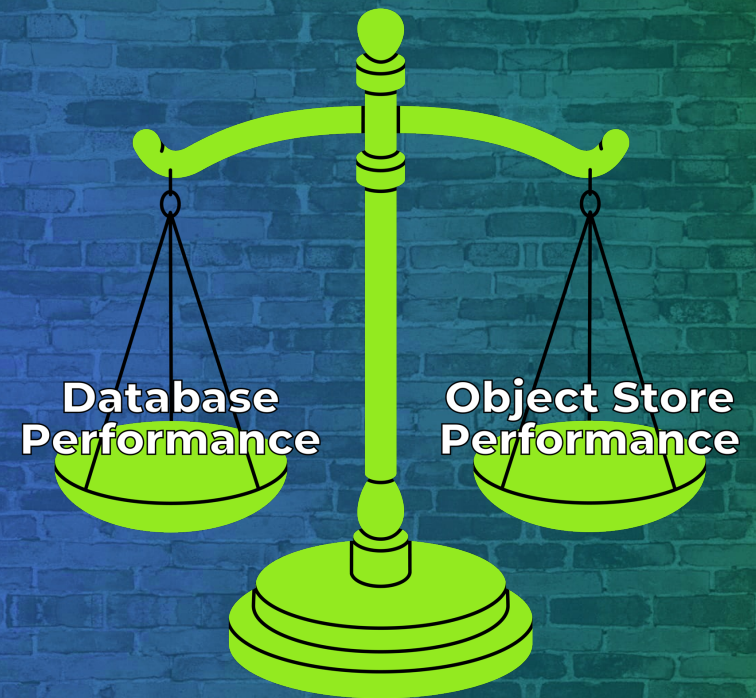
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Designing for Scalability

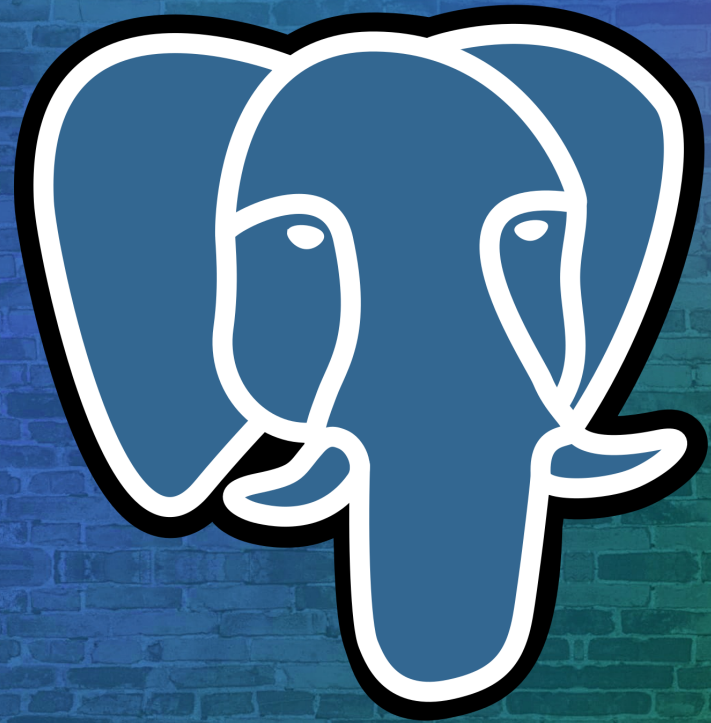
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PostgreSQL

Hardware and Design

- Always external, always Linux
- Hardware (50 proxies): 16 CPU / 64 GB RAM
- Hardware (150 proxies): 24 CPU / 96 GB RAM
- But proxies not the true metric
- Run **Set-VBOPSQLDatabaseLimits** after install & as part of upgrade procedures
- Maximum version: 15.9 or 16.4
- Do not share with VBR (VCSP VEX exception)



NATS

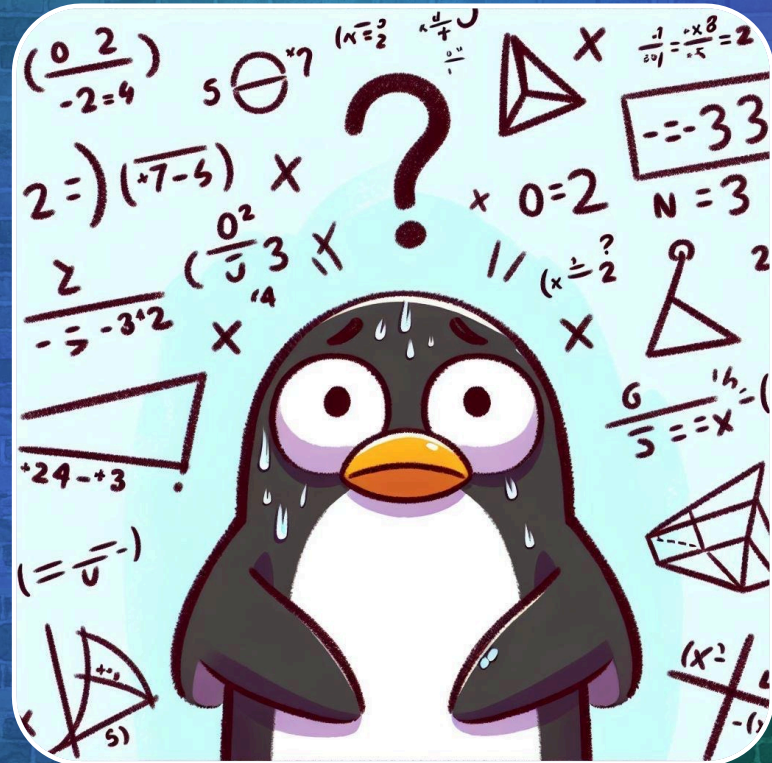
Hardware and Design

- Always External
- Hardware: 8 CPU / 32 GB RAM
- Single node for speed
- Consider containerizing for management
 - How To: <https://tinyurl.com/NatsOnPodman>
- Client Advertise if shared container host



Proxies

- Keep all components close
- Now supports Ubuntu 24.04, RHEL 9.4 in 8.1
- Cloud Instances
 - Azure D or F type instances
 - AWS M series instances
- 1:4 CPU:RAM ratio
 - 4 4x16 proxies more effective than 2 8x32



Pools



- Up to 3 pools, 50 proxies per pool
- Past 50 dedicated management proxy
- Dedicate 1 pool to onboarding orgs
- Start|Get|Stop-VBORepositoryOwnerChangeSession
- Build in mass but cost optimize by powering down

Backup Job Optimization

Discovery

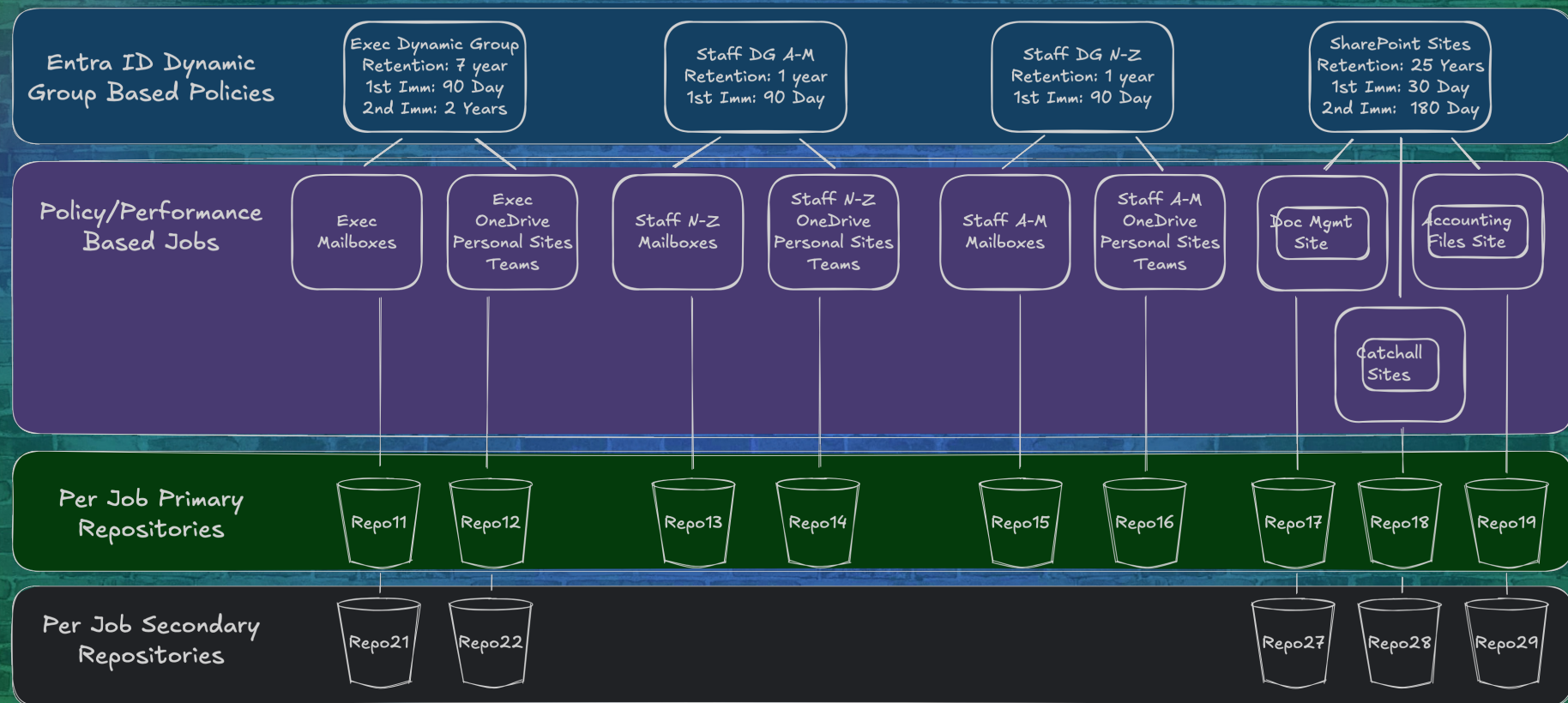
- Gathering Using Data
 - Microsoft Admin Center Usage Reports
 - <https://calculator.veeam.com/vb365>
- Be Aware of
 - Multi-geo organizations
 - Large Sharepoint sites
 - Compliance based retention data units
 - Unlicensed users/groups (shared, group, etc.)
- Design for very conservative splitting, organizations will grow/change

Implementation

- Split Organizations by
 - Retention Policies
 - Exchange / Sharepoint
 - 5000 users
- Add 2nd App Registration for heavy Sharepoint
- Segregate large Sites

Backup Job Optimization

In Practice



SaaS FTW!

When In Doubt, Get An Expert



Veeam Cloud & Service Provider (VCSP) Program

- Diverse marketplace of providers and services to provide a tailored solution
- Unlimited, immutable storage available
- Offload expertise to a Veeam Service Provider



Veeam Data Cloud *for Microsoft 365*

- When first party is a hard requirement
- Infrastructure & updates managed by Veeam
- Unlimited storage included

Thank You.

Slides and Materials Available At:

https://tinyurl.com/Architecting_VB365_v8